Analyzing Permutation Wordle

Family Meeting 4/13/26

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Permutation Wordle

1. I'm thinking of $\pi \in \mathcal{S}_n$
2. Your goal: guess π in as few guesses as possible
3. After each guess, I will tell you which indices (if any) are correct
4. The game ends when you have guessed my permutation

Example

12345

Example

1**2**345

Example

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52134

Example

1**2**345

5**2**1**3**4

Example

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42531

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$$\gamma_1 = \quad 1\mathbf{2}345$$

$$\gamma_2 = \quad 5\mathbf{2}1\mathbf{3}4$$

$$\gamma_3 = \quad \mathbf{42531}$$

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$$\gamma_1 = \quad 1\mathbf{2}345 \qquad \begin{cases} \mathcal{I}_1 = \{1,3,4,5\} \\ \mathcal{J}_1 = \{2\} \end{cases}$$

$$\gamma_2 = \quad 5\mathbf{2}1\mathbf{3}4 \qquad \begin{cases} \mathcal{I}_2 = \{1,3,5\} \\ \mathcal{J}_2 = \{2,4\} \end{cases}$$

$$\gamma_3 = \quad \mathbf{42531} \qquad \begin{cases} \mathcal{I}_3 = \emptyset \\ \mathcal{J}_3 = \{1,2,3,4,5\} \end{cases}$$

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Alt: *CS* has the smallest average number of guesses for $\pi \in \mathcal{S}_n$

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Note: There is rigidity! Easier to teach a computer, but limiting.

Cyclic Shift & Other Strategies

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$CS := [(1), (21), (231), \ldots, (2\ 3\ \ \ldots\ \ n\ 1)]$
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Later: $S := [D_1, D_2, D_3, \ldots, D_n]$ where $D_i \in Der(i)$

Generating Functions

$f_S(x) = \sum_{m \geq 1} a_m x^m$ where $a_m = \#\pi \in \mathcal{S}_n$ guessed in m guesses by strategy S

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Remarks:

- Maximum degree not fixed (bounded?)
- Coeffs of $f_{CS}(x)$ are Eulerian numbers! (Kutin-Smithline)
- For any S, $[x^1] f_S(x) = 1$ (trivial perm.)
- For any S, $[x^2] f_S(x)$ is fixed (let's prove it!)
- $[x^3] f_{CS}(x) > [x^3] f_S(x)$ for any other S
  $\hookrightarrow$ By induction! See Hiveley 2025, Section 4

One (Mandatory) Proof

Theorem

For all strategies S of length n, $[x^2]f_S(x) = A(n, 1)$

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- If $|\mathcal{J}_1| = k$, then there are $\binom{n}{k}$ ways to select correct entries
- $[x^2]f_S(x) = \#\mathcal{J}_1 = \sum_{k=0}^{n-2} \binom{n}{k} = A(n,1)$ (A000295)

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Example: For CS with $n = 6$:

123456	$\mathcal{J}_1 = \emptyset$
612345	$\mathcal{J}_2 = \{1, 2, 5, 6\}$
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Count $\mathcal{J}_2 \subset [n]$ s.t.:

- $|\mathcal{J}_2| \leq n - 2$ (cannot be n or $n - 1$ to end in 3 guesses)
- For CS: $\mathcal{J}_2 \neq \{i, i + 1\}$ for any $i \pmod{n}$
- In general: $\mathcal{J}_2 \neq \{j, k\}$ where $\pi[j] = i$ and $\pi[k] = j$

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Duplicated information $\implies$ BAD!

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Observations:

- Clearly can happen for non-cyclic derangements.
- True whenever cycle lengths pair off (see Hiveley 2026, Theorem 6)

Duplication (cont.)

Ex: $S = [(1), (21), (231), (2143), (34521)]$ w/ $\pi = 41523$

$$\begin{aligned} \gamma_1 &= 12345 & \mathcal{J}_1 &= \emptyset \\ \gamma_2 &= 54123 & \mathcal{J}_2 &= \{4, 5\} \\ \gamma_3 &= 15423 & \mathcal{J}_3 &= \{4, 5\} \\ \gamma_4 &= 41523 & \mathcal{J}_4 &= \{1, 2, 3, 4, 5\} \end{aligned}$$

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Intuitively, sure! Corollary of Kutin & Smithline

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Theorem

Every other strategy does

Proof by nasty constructive algorithms (see Hiveley 2026, and next slides)

Duplication Algorithm

Def. The displacement vector $d(\pi)$ of $\pi \in S_n$ is the list of length n s.t.

$$d(\pi)[i] := \pi[i] - i \text{ for all } 1 \leq i \leq n$$

i.e., how far away an entry is from "home" (fixed position)
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  - Later extend to *all* strategies by finding "first" inductive component

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- Special Cases:
  - $d(\pi)$ has only 1's and $(n-1)$'s (cyclic shift, or derangement)
  - $d(\pi)$ is all 2's

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In summary: need a new approach ...

What's Next?

- Coefficients of x^k for $k \geq 4$
  $\hookrightarrow$ Computationally difficult. A lot of case work using old approach, and excedance framework does not easily generalize
- Analysis of proposed "worst" strategy:
  $CSL = [(1), (21), (231), \ldots, \underbrace{(n\ 1\ 2\ \ldots\ n-1)}_{\text{left shifting}}]$

★ New collaborators???
  $\hookrightarrow$ Want: suboptimal duplication proves the conjecture

Thanks & Questions